PRACTICAL 8

Measures and Calculated Columns

Subject: Data Visualization for Decision Making

Objective: Create DAX measures and calculated columns to perform business calculations.

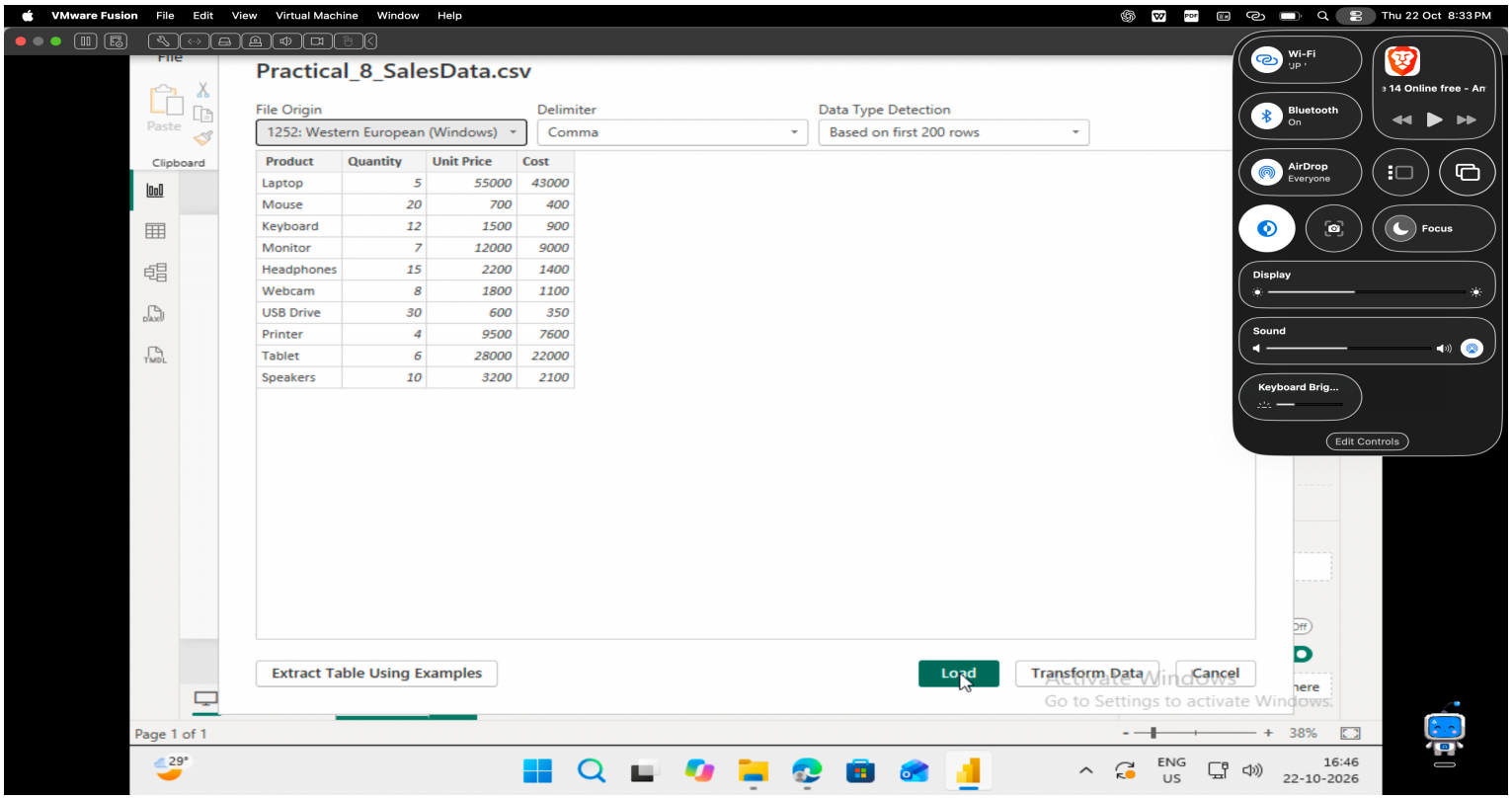
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Course	M. SC.IT (Data Science)
Semester	1
Course Code	P91A3DVD
Academic Year	2026–27

Practical work: creating row-level calculated columns and filter-aware DAX measures in Power BI Desktop.

Procedure

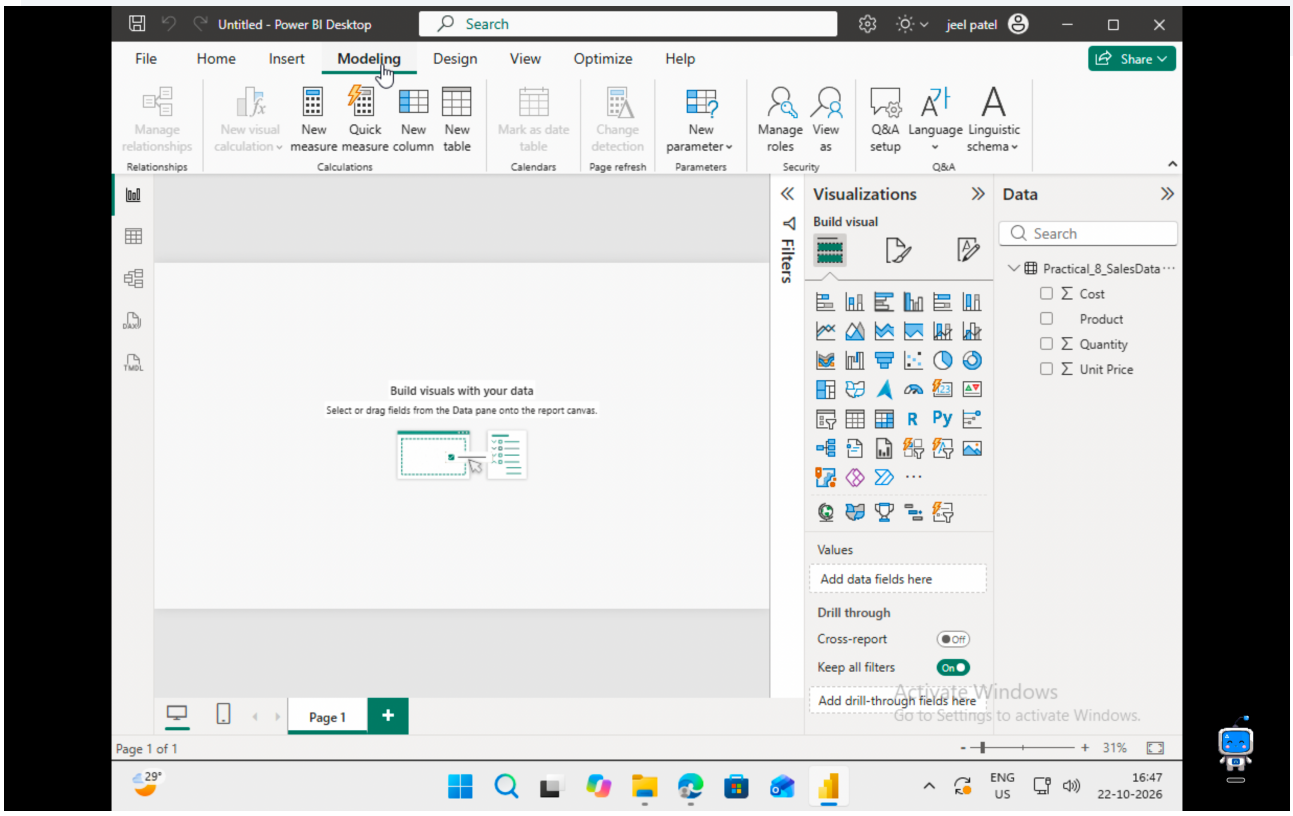
Sample data used

Create or import a table named **SalesData** with Product, Quantity, Unit Price, and Cost columns. The example below uses illustrative sample values.



Step 1: Create a calculated column for total sales

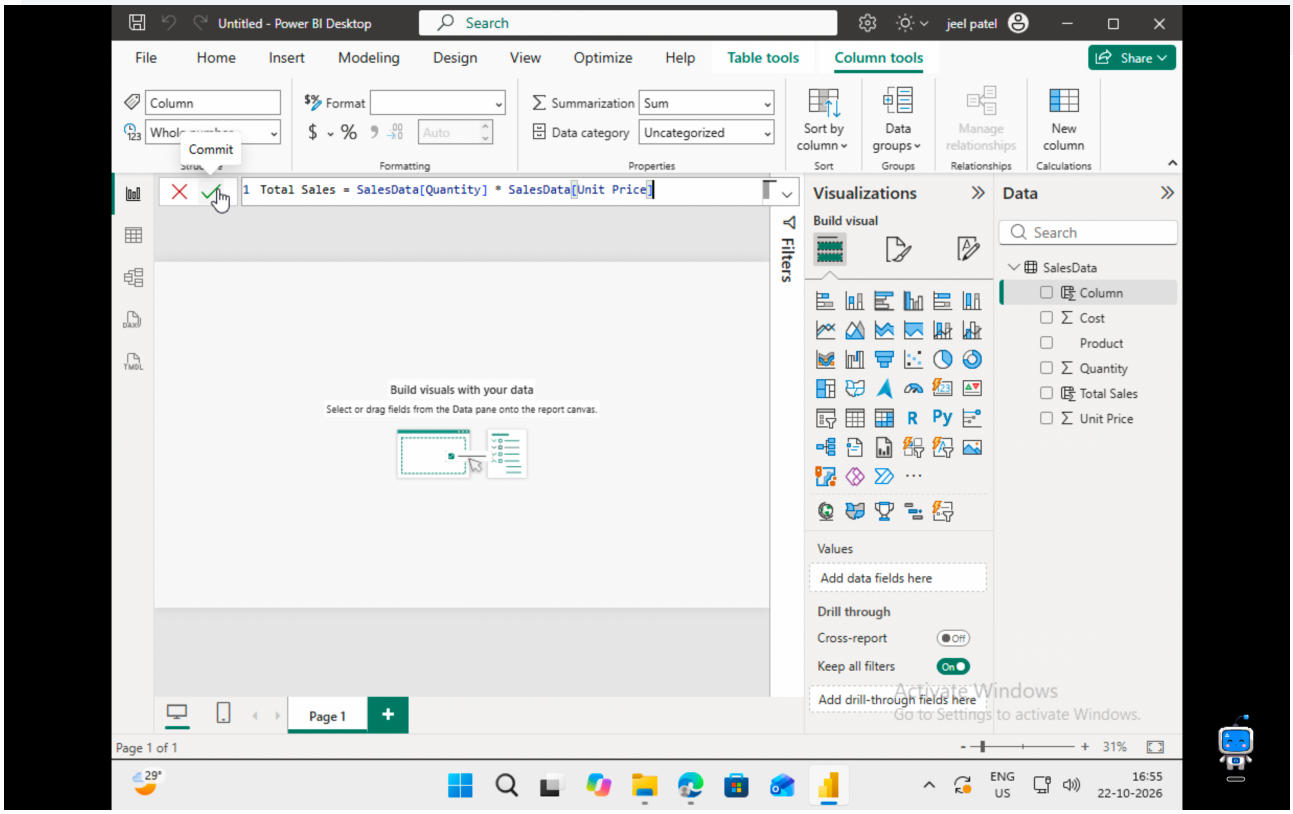
In Power BI Desktop, select the SalesData table, choose **Modeling > New column**, and enter the DAX formula. A calculated column evaluates the expression for each row.

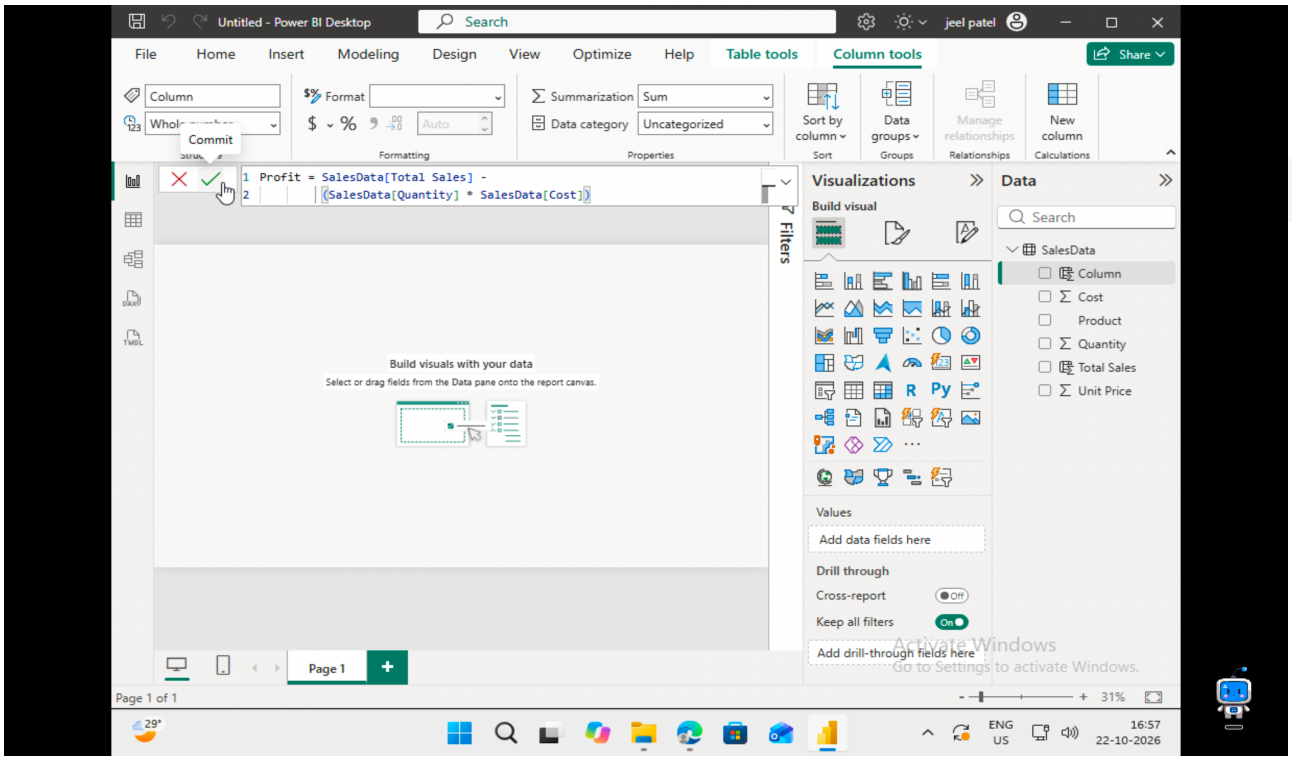


Note: *Ensure Quantity and Unit Price are numeric data types.*

Step 2: Calculate profit for each row

Add another calculated column to calculate the profit amount for each product row. This example treats Cost as a per-unit cost.

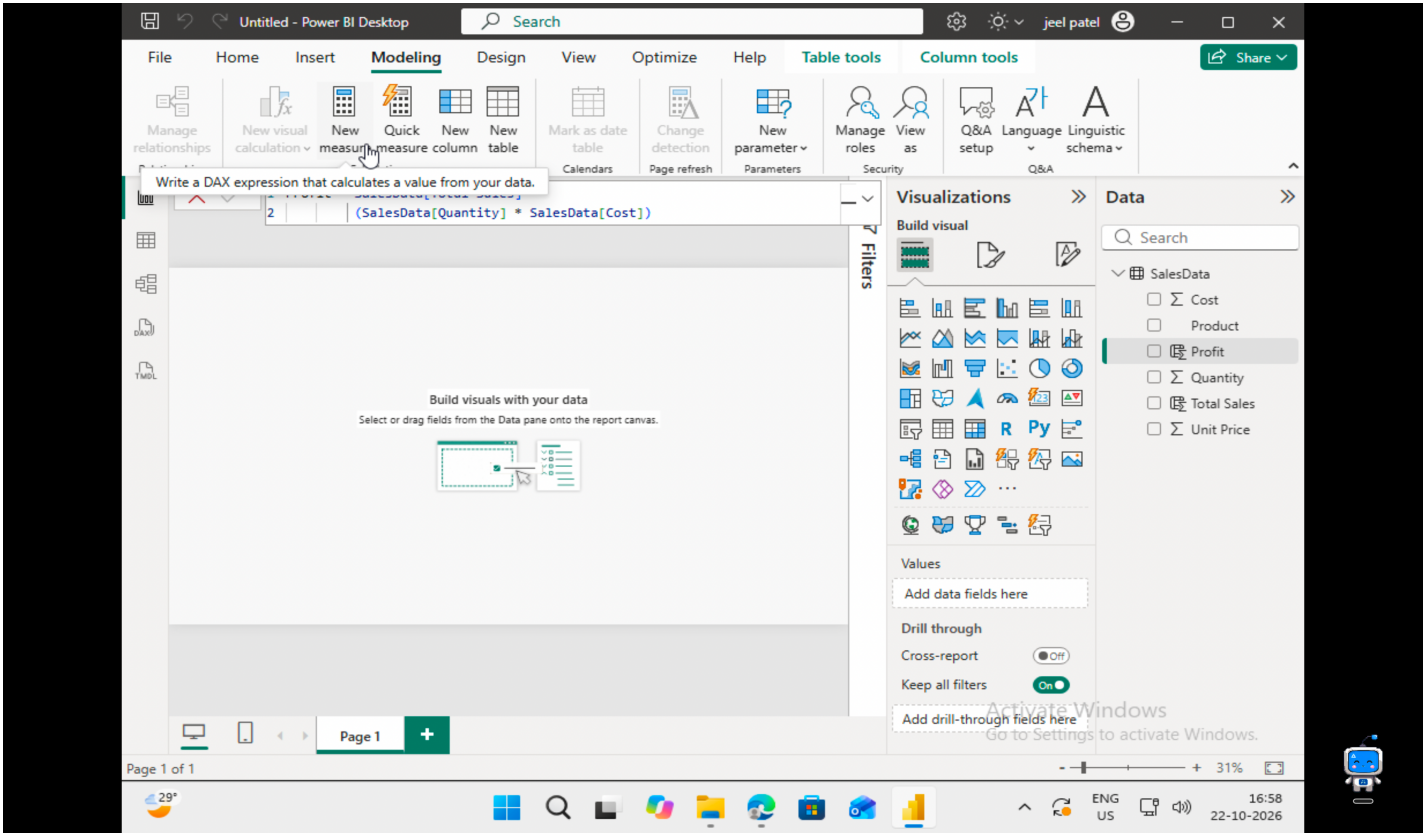




Note: *If Cost in your dataset is already a row total, use Total Sales - Cost instead.*

Step 3: Create a total revenue measure

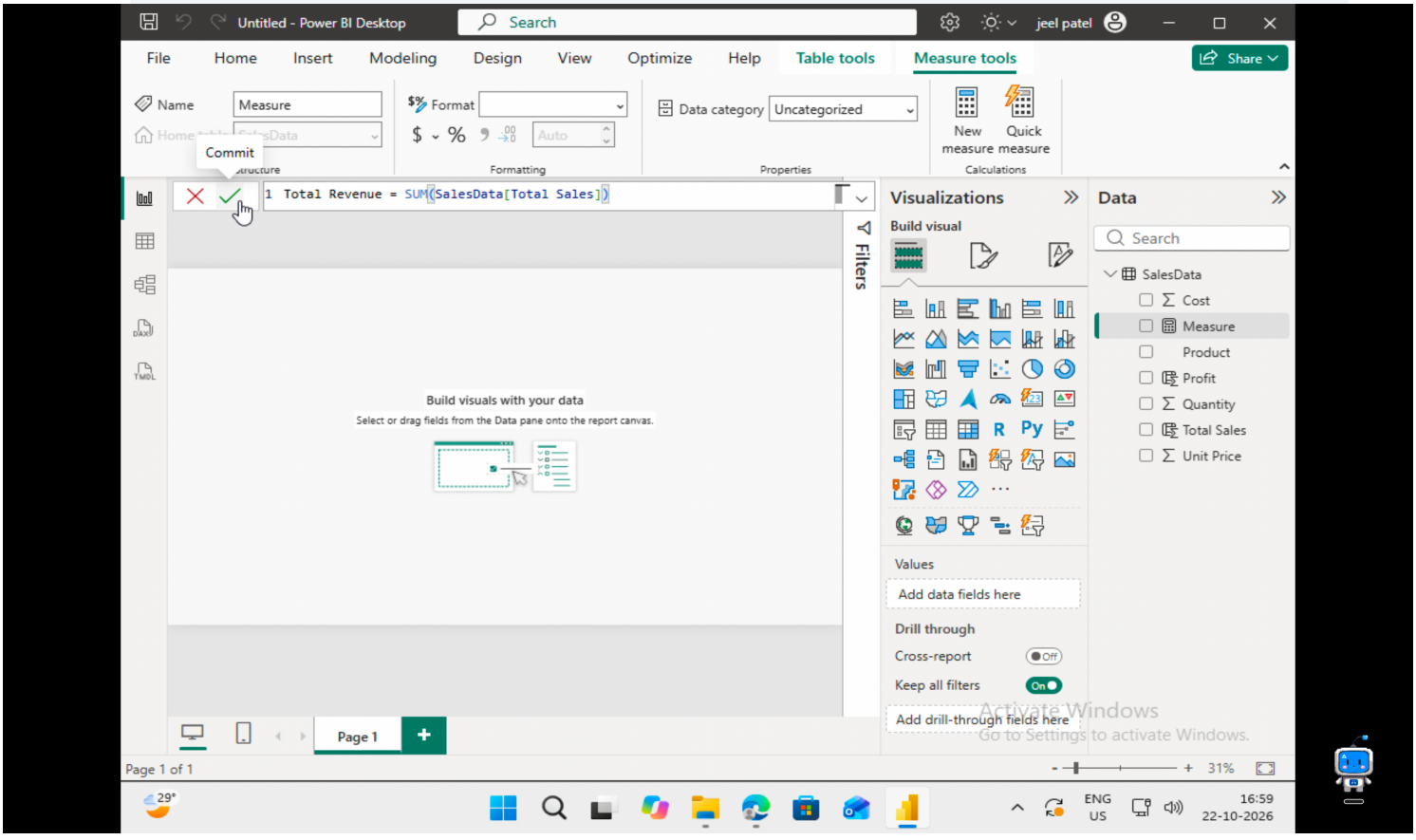
Select **Modeling > New measure** and create a measure that sums the Total Sales column. Measures recalculate according to the current report filters and visual context.



Note: *Measures are commonly used in cards, tables, and charts.*

Step 4: Create total profit, profit margin & total revenue measures

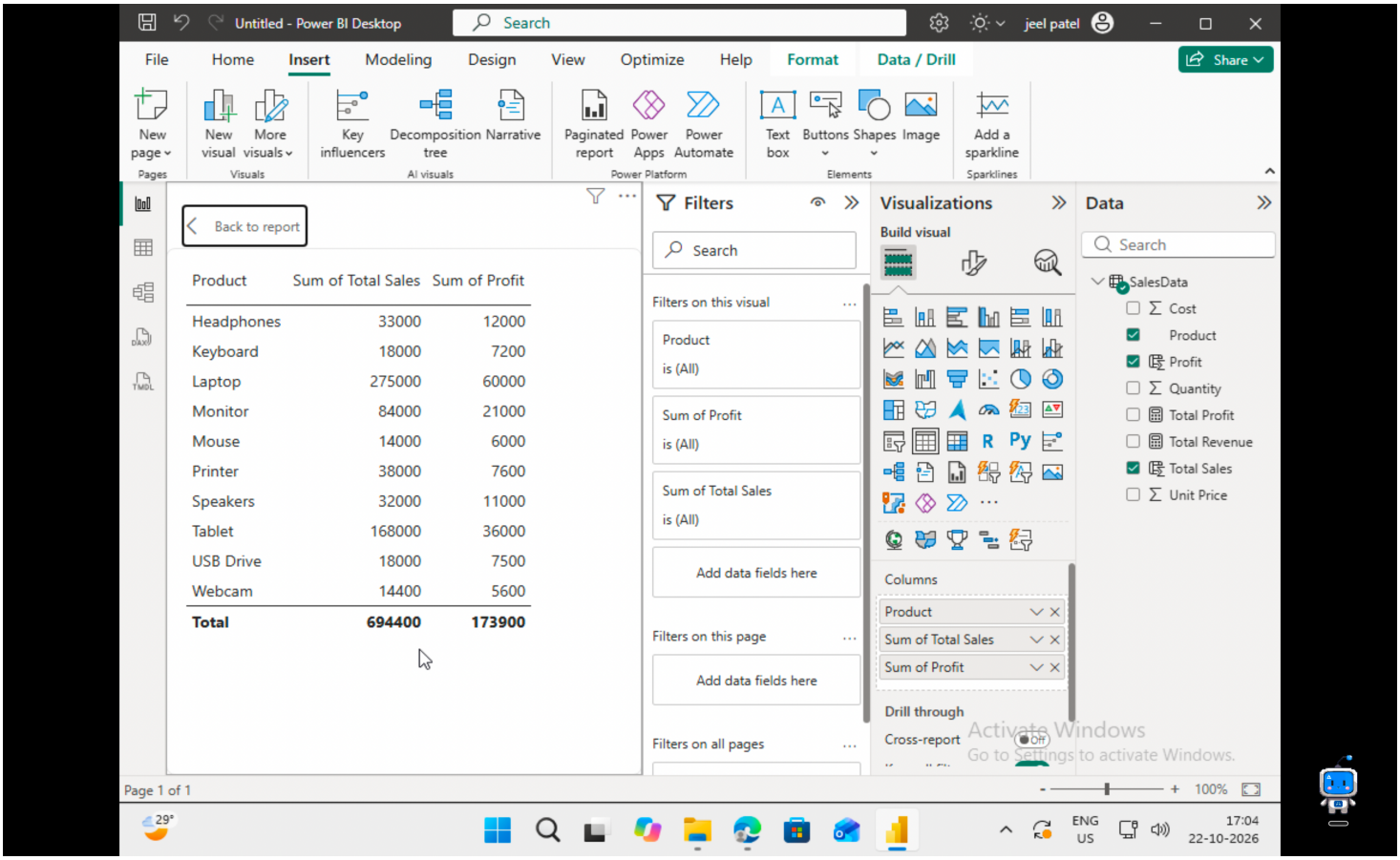
Create measures for total profit and profit margin. DIVIDE safely handles a zero or blank denominator using the alternate result provided.



Note: *Format Profit Margin % as Percentage from the Measure tools ribbon.*

Step 5: Validate calculations in a table visual

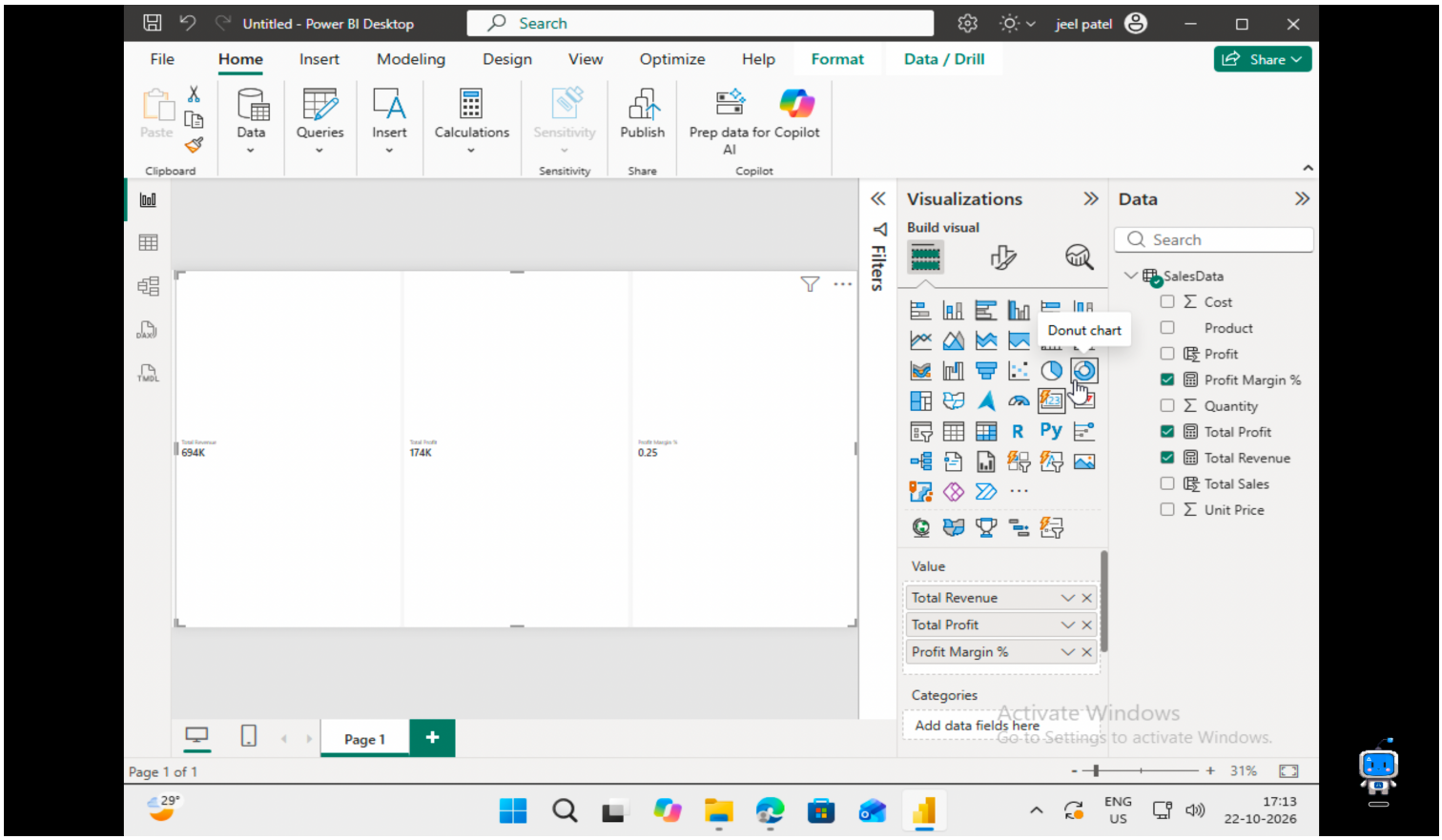
Insert a Table visual and add Product, Total Sales, and Profit. Check that row-level calculations are sensible and that totals aggregate correctly.



Note: *Displayed totals depend on the exact data and any filters applied.*

Step 6: Review summary measures and save

Add Card visuals for Total Revenue, Total Profit, and Profit Margin %. Verify the results, then save the report as a Power BI Desktop (.pbix) file.



Result

DAX calculated columns and measures were created to perform business calculations in Power BI. The calculated columns derive Total Sales and Profit for each row, while measures calculate Total Revenue, Total Profit, and Profit Margin % according to the report's filter context.

Conclusion: Calculated columns are useful for row-by-row values stored in the data model, whereas measures are suited to dynamic aggregations used in report visuals.